LESSON PLAN

PISTON ENGINE (15)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	03
Total No. of Lesson Plans	6	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Principles & operations of Super chargers, description and operations.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Principles & operations of Super chargers, description and operations.
5. Specific Objectives	15.1.2.1 Trainees will be able to describe terminology used for piston engine.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of supercharger drive mechanism and airflow path; Safety: Ensure engine is off and cool; Supervision: Instructor monitors trainee interaction with diagrams.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a supercharger?
(b) How does a supercharger differ from a turbocharger?
(c) List three types of supercharger drives.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition and basic principle of supercharging • Description of centrifugal supercharger components • Description of positive displacement supercharger types • Operation of gear-driven supercharger systems • Operation of belt-driven supercharger systems	L & Ds	PPT, IB	15 min
(b)	• Principles of pressure ratio and boost generation • Operation of variable vane superchargers • Operation of Roots blower mechanisms • Operation of centrifugal impeller dynamics • Impact of supercharging on engine intake charge	L & Ds	PPT, PDF	20 min
(c)	• Maintenance requirements for supercharger units • Troubleshooting common supercharger failures • Inspection of drive belts and gears • Lubrication points for supercharger assemblies • Cooling system integration with superchargers	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the operational difference between a Roots blower and a centrifugal supercharger.
(b) Describe the role of the supercharger in increasing engine power output.

12. Home Assignment and Follow Up	Review LP 02 syllabus and identify how engine terminology relates to supercharger operation.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Terminology used i.e. BDC TDC Stoke length, clearance volume, swept volume, valve lead, valve lag, ineffective crank angle	Refer to the master syllabus reference list	LP 04

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min